

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, TypeScript, C++, Java, SQL

Tools and Frameworks: PyTorch, OpenCV, LangChain, LangSmith, Pinecone, REST(Django, FastAPI), React, Next.js, PostgreSQL, PyTest, AWS, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD), Git, Linux

Areas of Interest and Coursework: Distributed System Design, AI & LLMs, Data Structure & Algorithms

INDUSTRY EXPERIENCE

Data AI Intern | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services for an AI-native **GRC platform** that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.
- Implemented the **RAG chat agent** with its React interface, marking each control fully, partially, or not satisfied and answering follow-up questions on the unmet ones, with every answer grounded in citations retrieved from the vector index, so no compliance verdict rests on **fabricated control text**.
- Built a priority-ordered **fallback across 5 embedding models**, re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising **evidence-backed control assessment accuracy from 64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an AI-powered **PRD review platform** in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.

Full Stack Developer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Architected a **self-serve quoting platform** that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from **~2 days to <1 min**.
- Built the **online ordering and booking platform** by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to Kafka workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across ~1,100 orders.
- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10× seasonal peaks and load-tested to **~50 quote requests/sec at <120 ms p99** with zero-downtime rolling deploys.

PROJECTS

When Pixels Talk Back | Watermarking EMR Data to analyse impact on CV models

- Fine-tuned **3 computer-vision models** (ResNet50, DenseNet169, MobileNetV2) for classification over COVID-19 lung CT scans via two-stage ImageNet transfer learning, to establish a baseline peaking at **93%** accuracy and **0.951 F1**.
- Embedded a **1024-bit blind fragile watermark** into the Region-of-Non-Interest of CT scans via key-scrambled LSB embedding of a SHA-512 fingerprint, enabling content authentication with near-lossless quality at **SSIM 0.999**.
- Benchmarked **3 watermarking algorithms** across all 3 models, showing the fragile scheme costs **0%** accuracy loss, versus 11.2% for robust embedding and ~1% for reversible embedding with a 300-bit payload.

Nothing Lost in Translation | Lexically constrained Sequence Generation

- Built a **Grid Beam Search decoder** that guarantees positive/negative lexical constraints in any autoregressive sequence generation, demonstrated on English-to-Russian NMT with a **15-point BLEU** gain (25.45 → 40.59).
- Reconstructed a beam search decoder from scratch, outperforming HuggingFace's stock `.generate()` by **1.45 BLEU**.